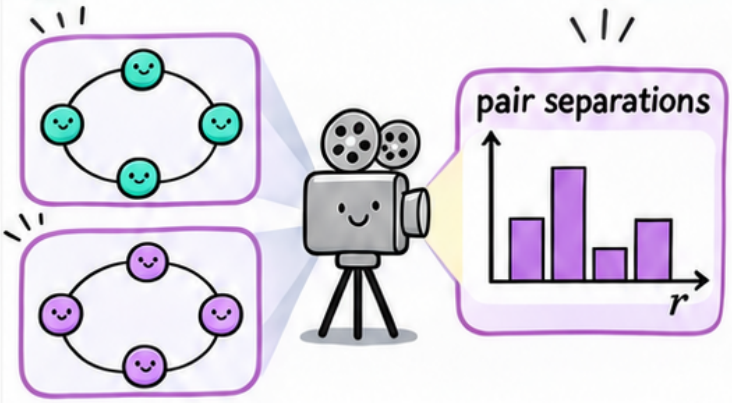
PAIR DATA ARE A SHADOW

The same distances can hide different motion.

1 What does a pair statistic remember?

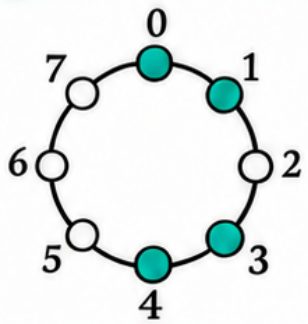


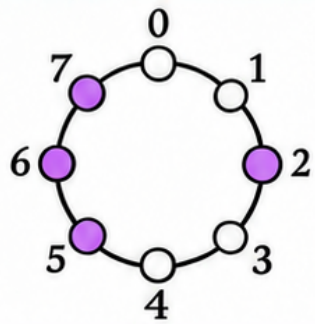
$$p_k = \sum_j \exp(ik\theta_j)$$

All $|p_k|^2$ together record all pair displacements.
But it forgets which pairs share a zero.

Unlabelled pair data ≠ a labelled distance matrix.

2 An exact eight-site example





$S = \{0, 1, 3, 4\}$ $C = S^c = \{2, 5, 6, 7\}$

Same six distances: 1, 1, 2, 3, 3, 4.

$$c_C(r) = 8 - 2 \cdot 4 + c_S(r) = c_S(r)$$

$c_S(r)$ *counts occupied pairs at displacement r.*

3 Let the zeros move

Heat s

$$\partial_s F = \partial_x^2 F$$

Each Fourier mode k decays by $\exp(-k^2 s)$.

τ = first collision time

S $\tau_S \approx 0.08863$

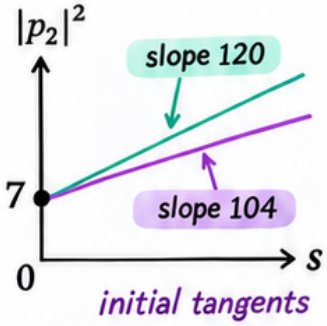
C $\tau_C \approx 0.11552$

Same full pair data. Different first collision.

Both examples are grid configurations.

4 The missing information is already cubic

Six points on twelve sites:
$A = \{0, 1, 2, 4, 7, 8\}$,
$B = A^c$



initial tangents

- Same pair data AND same sorted gaps.
- $T = \mathrm{Re}(p_1^2 \overline{p_2}) = +2$ on A, -2 on B.
- $|p_2|^2$ starts at 7; initial slopes: 120 vs 104.

Heat detects how several neighbours act on one zero.

This six-point perturbation changes a protected cubic.

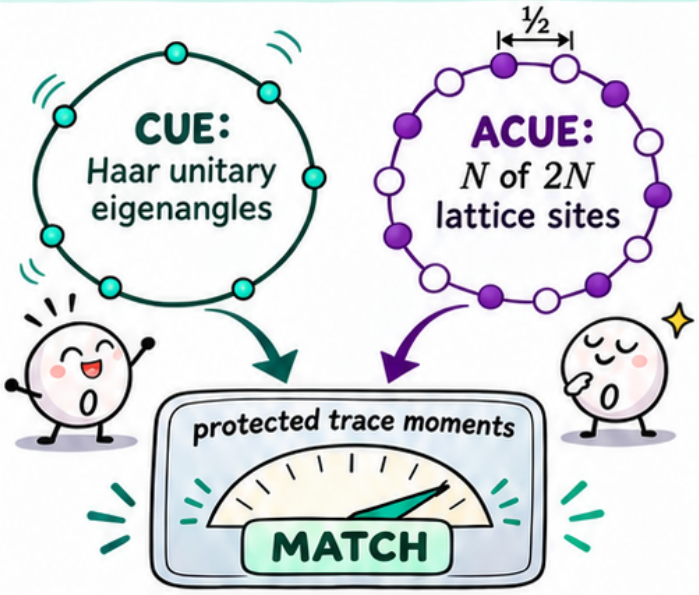
Counterexample to reconstruction from pair data.

Finite-model results from the Tao meeting note, 5 November 2026. Not a counterexample to the zeta/GUE conjecture.

TWO FLOWS. ONE INFORMATION BARRIER.

What the counterexamples teach us about zeta zeros.

5 Tao's motivation: a convincing imitation



- ACUE uses Vandermonde-squared weights.
- Its gaps are multiples of half the mean spacing.
- Yet a whole protected range of moments agrees exactly.

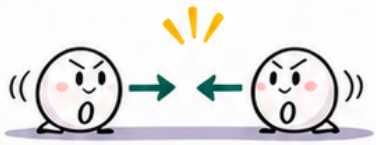
Matching accessible statistics need not identify the process.

6 Which flow are we using?

SCALAR HEAT

$$\partial_s F = \partial_x^2 F$$

- Deterministic, attractive zero motion.
- First collisions reveal missing information.



DYSON BROWNIAN MOTION

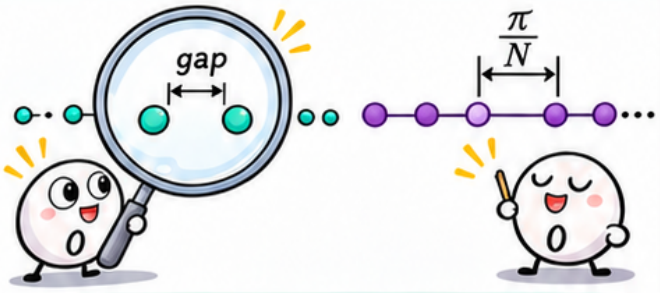
- Noise + repulsive drift.
- Protected multitime moments *STILL* match.



Shared across all times: total positive weight $\le N$ and total negative weight $\le N$.

Adding dynamics does not automatically break the protection.

7 A first collision is an extreme



Finite CUE: τ has scale $N^{-8/3}$

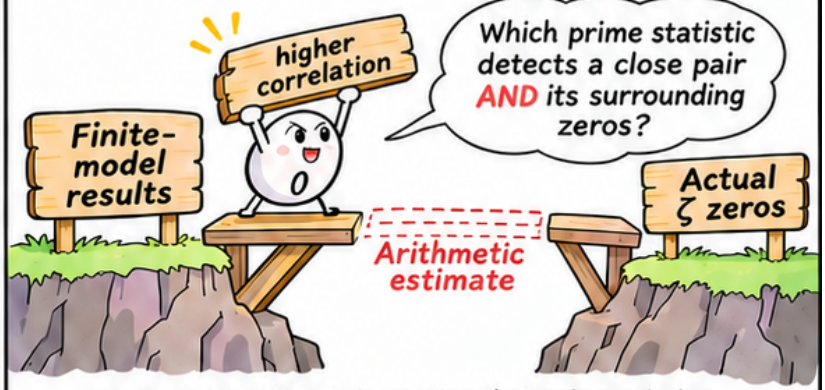
ACUE: $\tau \ge \frac{\pi^2}{8N^2}$

- For an isolated close pair: $\tau \approx \text{gap}^2/8$.
- Fixed-window convergence does not control the rarest gap.
- CUE scale is probabilistic. The ACUE bound is not an equality.

A local statistical law is not yet a global extreme-value law.

8 The missing bridge to real zeta zeros

primes → higher zero correlations → collision control



- We have identified information that pair statistics can miss.
- To reach zeta, we must estimate that information at the right scales.

The finite periodic heat flow is not itself the zeta/Newman flow.

Proved finite-model obstruction. Open arithmetic bridge.

Based on the Tao meeting note and internally checked project proofs. No proof or disproof of RH or the Montgomery–Dyson conjecture.